

Modelling the enablers of sustainability in higher education institutions

Higher
education
institutions

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Received 24 July 2019
Revised 11 July 2020
28 September 2020
Accepted 25 December 2020

Abstract

Purpose – The purpose of this paper is to explore and encapsulate the enablers that can facilitate education for sustainable development in higher education (HE). The study also aims to understand the interdependence between the enablers.

Design/methodology/approach – The study adopts the total interpretive structural modelling approach to comprehend the interaction and transitivity between the enablers. Cross-impact multiplication matrix analysis was applied to rank the enablers and classify them on the basis of the driving and dependence power into dependent, autonomous, independent and linkage enablers.

Findings – An extensive literature review and expert opinion helped in identifying 10 enablers that can promote sustainability in higher education. The structural model revealed government policies, media, accreditation/sustainability audit, sustainability leadership and institutional commitment as the crucial enablers that can drive sustainability and activate the enablers with high dependence and low driving power.

Practical implications – The results of this study will assist the policymakers and management of universities and colleges in understanding important factors that can facilitate sustainability in higher education. Universities and colleges to incorporate sustainability in their system need to transform not only the core higher education activities of learning and teaching, research and engagement, also the way the colleges operate its culture, governance, structure and how it supports the staff and students.

Originality/value – So far, research on sustainability in higher education has looked into each factor in isolation. This research provides a comprehensive view of the factors and has attempted to establish a multidirectional interplay between factors facilitating sustainability in higher education (SHE).

Keywords Management, Education, Modelling, Sustainability, Higher education, Education for sustainable development, Total interpretive structural modelling

Paper type Research paper

1. Introduction

Though the economic growth has contributed to increasing national income and better quality of life, but it has also resulted in degradation of the environment, growing social inequality and a widening gap between the haves and the have-nots (Patnaik, 2018). These unsustainable practices, if ignored, can lead to irreversible damages. The Brundtland (1987) report released by the World Commission on Environment and Development was the first to introduce the sustainable development concept. The report defined sustainable development as the development that meets the needs of the present without compromising on the ability of future generations to meet their needs. The report triggered the international awareness and concern for development that is environment friendly, economically viable and socially just for the present and future generations (Sriskandarajah *et al.*, 2010). The United Nations Conference on Environment and Development held in Rio de Janeiro in 1992 emphasized the need for incorporating sustainability in the educational system (Figueiró and Raufflet, 2015) as education is believed to influence the thoughts and actions of people (DESD, 2005). To



promote sustainable living, UNESCO proclaimed 2005–2014 as the decade of education for sustainable development (DESD) with the objective of integrating the values, principles and practices of sustainable development into all aspects of education and learning (Wals, 2014).

To grapple with complicated societal issues, the current generation needs to be equipped with the skills and competencies that enhance their critical thinking, problem-solving abilities and reflexivity (Remington-Doucette and Musgrove, 2015; Tarrant and Thiele, 2016). Higher education institutions (HEIs) are the hubs of knowledge creation and can serve as a powerful means of imparting sustainable living. Higher education has also been identified as a key player in achieving sustainable development goals (SDGs), especially the fourth goal of imparting equitable and quality education. In pursuit of imparting education that enables the present generation to commit to a sustainable life, HEIs have been reorienting their institutional practices and policies. The DESD and the initiatives by international organizations have resulted in a number of colleges and universities signing declarations, committing to integrating sustainability into their system (Lozano *et al.*, 2015). To understand how far these declarations and charters have been successful in helping colleges and universities adopt sustainable practices, a number of research studies, both conceptual and empirical, have been conducted in the past. There have been a myriad of studies on the ways of incorporating sustainability principles into the curriculum (Dagiliūtė and Liobikienė, 2015; Chhokar, 2010). Studies have pointed out the need for a change in the pedagogical approach for teaching a multidimensional and complicated concept like sustainability (Rodriguez-Andara *et al.*, 2018; Fini *et al.*, 2018; Restrepo *et al.*, 2017). Mintz and Tal (2018) posited on linking sustainability course with appropriate pedagogy for enhanced sustainability learning outcome in students. Lozano *et al.* (2015) emphasized on incorporating and institutionalizing sustainability into the processes and practices of HEIs. Menon and Suresh (2020a) studied the practices adopted by colleges and universities across the globe in incorporating sustainability in education and research (Vemury *et al.*, 2018; Beringer and Adom̃ent, 2008), campus operations (Murray, 2018; Mohammadalizadehkorde and Weaver, 2018) and outreach programmes (Cho, 2017). Studies have also pointed out at the importance of leadership (Scott *et al.*, 2012), the role of government (Fiselier *et al.*, 2018; Su and Chang, 2010), teachers' skills (Biasutti *et al.*, 2018), assessment and reporting (Yarime and Tanaka, 2012; Saadatian *et al.*, 2011) and institutional commitment (Velazquez *et al.*, 2006) in driving sustainability integration in higher education.

Although there have been a number of studies that have pointed out the factors that can drive sustainability in higher education, a comprehensive and holistic conceptual framework that identifies all the enablers and explains the contextual relationship between the variables has not been explored so far, and this became the motivation for this study. To fill the research gap, the study proposes to use the total interpretive structural modelling (TISM), which is a widely applied qualitative technique to analyse the interplay between the enablers and help develop a structural model depicting the multi-directional (direct and transitive) links between the variables. The TISM model also allows capturing the experts' opinion on logical interpretation between the variables (Vinodh, 2020). Cross-impact multiplication matrix (MICMAC) analysis helps rank the variables on the basis of their driving and dependence power.

The main objectives of this paper are:

- To identify the enablers that can facilitate education for sustainability in HEIs in the Indian context;
- To establish relationship and interdependence between enablers with the help of the TISM technique; and
- To analyse and rank the driving and dependence power of the enablers affecting education for sustainability using MICMAC analysis.

2. Literature review

The literature review included research papers on enablers that would facilitate sustainability in higher education and the papers that have used the TISM technique.

2.1 Literature on sustainability enablers

2.1.1 Sustainability incorporated curriculum. Sustainability incorporated curriculum refers to the lessons and academic content on sustainability taught in colleges and universities. [Ceulemans and De Prins \(2010\)](#) suggested two ways of incorporating sustainability: horizontal integration and vertical integration. While horizontal integration involves weaving courses on sustainability across the curricula of different disciplines ([Dagiliūtė and Liobikienė, 2015](#)), vertical integration includes exclusive courses on sustainability education. Principles of sustainability may be incorporated into the curriculum in a shallow way by including a topic, case material or a module on sustainable development in the existing programme ([Roure et al., 2018](#)). An elective course on sustainability, a degree course with a specialization in sustainable development or a new programme exclusively on sustainability are some of the other ways that HEIs have adopted in integrating sustainability into the curriculum ([Chhokar, 2010](#); [Maruna et al., 2018](#)). Some authors argued on the ineffectiveness of an add-on course and recommended that sustainability should be made an integral part of the system ([Sidiropoulos, 2018](#); [Savelyeva and McKenna, 2011](#)). [Ferrer-Balas et al. \(2010\)](#) advocated an incremental and participative approach to incorporating sustainability. [Stubbs \(2013\)](#) recommended knowledge, skills and attitude approach to design sustainability curricula. Sustainability focused curriculum should enhance cognition, develop emotions and promote experiential, reflective and collaborative learning ([Aktas, 2015](#); [Shrivastava, 2010](#); [Sipos et al., 2008](#); [Balsiger et al., 2017](#); [Piasentin and Roberts, 2018](#)). The multidimensional nature of sustainability requires integrating multiple perspectives and disciplinary expertise to solve sustainability issues ([Lozano, 2010](#)). A number of researchers have advocated interdisciplinary and multidisciplinary courses ([Savelyeva and McKenna, 2011](#); [Fisher and McAdams, 2015](#); [Juárez-Nájera et al., 2006](#); [Lidgren et al., 2006](#)). [Kurland et al. \(2010\)](#) and [Widener et al. \(2016\)](#) studied interdisciplinary courses taught at two different universities. Their study revealed interdisciplinary courses are better than disciplinary courses as these courses are taught by faculty members from different disciplines and this contributes to an integrated and holistic conceptualization. [Wu and Shen \(2016\)](#) observed that in the period 2005–2009, the curriculum of business schools mostly focused on environmental issues. The pressing need to include the triple bottom dimensions ([Elkington, 1997](#)) made institutions reorient their curriculum to integrate the environmental, economic and equity dimensions ([Rampasso et al., 2018](#)). [Remington-Doucette et al. \(2013\)](#) and [Mochizuki and Fadeeva \(2010\)](#) asserted that educational programmes must aim at enhancing the competencies of students in not only understanding the interconnectedness between humans and nature but also the conflicts and trade-offs between the social, economic and environmental dimensions. [Rusinko \(2010\)](#) and [Pandey and Vedak \(2009\)](#) developed matrices and models for colleges and universities struggling with integrating sustainability into their curriculum. [Barrett et al. \(2017\)](#) outlined the threshold concepts that can help in curriculum design. [Perera and Hewege \(2016\)](#) emphasized on identifying the learning gaps before making changes to the curriculum and [Wyness and Sterling \(2015\)](#) explained the importance and relevance of curriculum review for identifying the gap between the current and desired sustainability education.

2.1.2 Pedagogy. Pedagogy is the teaching strategy that can transform students “perspectives and behaviour by making them critically reflect on their value system” ([Viegas, 2016](#)). [Fisher and McAdams \(2015\)](#) emphasized that it is not the number of courses

offered, but the quality and the innovative ways of engaging students that helps them understand the concept and makes all the difference in their behavioural outcome. For students to reflect on their values and attitude, colleges and universities should integrate transdisciplinary learning (Shephard, 2008; Sterling and Thomas, 2006). Learning culture that blends formal and informal learning can develop meta-cognitive competencies that would enable students to cope up with the complex sustainability issues (Scott, 2015; Meyer, 2016; Barth *et al.*, 2007). Inductive teaching-learning that is learner-centred, interactive and participative (Kerekes and Wetzker, 2013; Seatter and Ceulemans, 2017; Bhatia and Bhatia, 2008) can facilitate understanding of a complex and ill-defined concept like sustainability. To understand the complexity and the interdisciplinary nature of sustainability issues, students have to live the experience and this is possible only by incorporating pedagogical tools that take students closer to reality (Sterling and Thomas, 2006). The case study is one methodology that brings the real world to the classrooms (Montiel *et al.*, 2018; Steiner and Laws, 2006) and judicious choice of transdisciplinary case studies can facilitate cognitive and affective learning among students, essential for understanding sustainability (Posch and Steiner, 2006; Steiner and Laws, 2006; Stauffacher *et al.*, 2006). Emancipatory pedagogy involving inquiry, problem or project-based learning encourages autonomy, accountability and creativity (Fini *et al.*, 2018; Restrepo *et al.*, 2017; Leal Filho *et al.*, 2016; Wiek *et al.*, 2014; Hallinger and Lu, 2011) compared to technical and practical pedagogies that involve passing on instructions, solving structured problems and rote memorization (MacVaugh and Norton, 2012). Some of the pedagogical approaches that provide students with experiential learning (Kolb and Kolb, 2005; Savage *et al.*, 2015) involve internships (Rodriguez-Andara *et al.*, 2018; Rios *et al.*, 2018), service learning (Albareda-Tiana *et al.*, 2018; Wang *et al.*, 2018; Martin, 2015), site visits (Kim *et al.*, 2018), outdoor adventure trips (Cusick, 2009) and work-based learning (Wall *et al.*, 2017). Role plays, simulation, discussions and debates (Gordon and Thomas, 2018; Shephard, 2008), social constructive pedagogy (Chandler and Teckchandani, 2015), concept mapping, participatory action research and interlinked/jigsaw teams that encourage cooperative peer-learning (Lozano *et al.*, 2017) can be effective pedagogical strategies for teaching sustainability. Innovative pedagogies such as aesthetic methods involving different art forms such as poems, drawing, paintings, photos, drama and movies are gaining popularity in teaching SHE (Ivanaj *et al.*, 2014; Keles, 2015; Österlind, 2018; Clemens and Hamakawa, 2010). According to Diamond and Irwin (2013) and Jääskelä *et al.* (2017), interactive platforms such as e-learning (Altomonte *et al.*, 2016), educational games, Lego serious play can be effective ways of engaging students in a playful and creative manner in solving sustainability issues (Roth *et al.*, 2015; Mercer *et al.*, 2017). Vemury *et al.* (2018) advocated multiple teaching-learning methodologies/pluralistic epistemologies in lieu of traditional positivist epistemologies effective for teaching sustainability (Christie *et al.*, 2013; Tarrant and Thiele, 2016).

2.1.3 Educators' skills and competencies. Falkenberg and Babiuk (2014) identified curriculum, pedagogy and teacher education as the three crucial areas of the educational process that need to be addressed for the successful incorporation of sustainability in the educational activities of higher education. A survey of the students of the pre-service teacher education programme revealed that although the curriculum includes environmental, social and economic sustainability issues, insights into these dimensions is superficial (Stir, 2006). A similar study was conducted by Falkenberg and Babiuk (2014) in the province of Manitoba in Canada. The study revealed lack of leadership, lack of clarity, resistance to the idea of sustainability and departmental silos as some of the hindrances to mainstreaming sustainability in teacher education programmes. Competency in incorporating principles, values and perspectives of sustainability into the academic courses and aligning the

delivery of the course with an appropriate and innovative pedagogical approach has become imperative (Biasutti *et al.*, 2018). Bertschy *et al.* (2013) compartmentalized competencies of an educator into three categories: the networking competency that enhances the educators' capability of understanding issues from a holistic perspective, the environing competency that helps educators plan for the future and the competency of transforming their attitudes and the education system by introducing new sustainable practices. Capacity building initiatives, faculty development programmes and learning communities (Nikolic *et al.*, 2017; Lozano-Garcia *et al.*, 2008; Natkin and Kolbe, 2016; Sibbel, 2009) can be effective in empowering the educators as collaborative peer learning enhances collegiality between faculty members and provides them with an opportunity to discuss and reflect on new teaching practices and other related matters outside the purview of their discipline (Biasutti *et al.*, 2018; Pedersen, 2017). Fien *et al.* (2000) pointed out that for educators to teach sustainability, it is important that the educators believe and practise sustainability in their personal life. Jaspas (2008) iterated faculty members who find the time and way to incorporate sustainability in their teaching-learning, personally find it valuable and meaningful.

2.1.4 Collaborations/partnerships. Collaboration can be defined as an arrangement in which different stakeholders (students, research scholars, academicians, research institutions, business leaders and community) come together and work in a non-hierarchical structure on a common sustainability goal. Lozano-Ros (2003) iterated collaboration and cooperation among faculty members and other stakeholders as the road to trans-disciplinarity. Collaboration within and between departments, partnership with stakeholders outside the campus (industry, research laboratories and international universities) and social collaborations (Bhatia and Bhatia, 2008; Moore, 2005; Holmberg and Larsson, 2017) are some of the collaborations that HEIs should nurture. The partnership between government, research institutions and private companies can promote inclusive innovations that are critical for countries that lack financial resources. HEIs and industry partnerships can strengthen research and development, gain expert advice, financial support and help in commercializing laboratory sustainable innovations (UNESCO, 2012). Internationalization of HEIs by collaborating with other universities abroad for research, academic exchange, technology transfer and cross-cultural links would benefit individuals, institutions and society (Vardhan and Shahul Hameed, 2012). Transnational collaborations and the use of digital technology have resulted in a fast and effective integration of knowledge across geographical boundaries. Caniglia *et al.* (2017) recommended HEIs to initiate and invest in transnational collaboration projects in sustainability to accelerate the learning process and solve local issues by connecting with experts globally. A Web survey of 14 HE networks conducted by Dlouha *et al.* (2018) revealed a wide range of diversity in the HE network activities. These networks carry out sustainability initiatives within and outside the institution. The networks support institutions in initiatives related to integrating sustainability in education and research, community programmes, campus operations, professional development, student networking and stimulating policy debates. Gadsby and Bullivant (2011) reiterated collaboration not only promotes global citizenship but it also motivates people to collectively face and solve sustainability challenges. The HEIs should design and implement an appraisal system and incentive structures that encourage collaborations (Yarime *et al.*, 2012).

2.1.5 Research and innovation. Research is the generation of new ideas. HEIs are the centre for knowledge creation. The complexity and uncertainty faced by society need to be addressed by developing new socially relevant knowledge and disseminating the same to the public and policymakers. To get global perspectives on the issues faced by society

today, researchers across the globe need to collaborate, deliberate, cooperate and solve the problems (Wickson *et al.*, 2006). Waas *et al.* (2010) observed that unlike conventional research practices that are usually restricted to the discipline of the researcher, sustainability issues require a holistic approach to problem-solving. The problems faced by society are real-world problems that call for a viable and practical research outcome that can be implemented (Bossio *et al.*, 2014). Such research projects require researchers from different disciplines to collaborate and integrate approaches, insights and methods of their specific discipline and synthesize these insights to construct a new methodology unique to the problem (Wickson *et al.*, 2006). Tormey *et al.* (2008) and Wooltorton *et al.* (2015) advocated action research as an effective way of collaborating and reflecting on the educational practices and approaches. Aktas (2015) believed student-educator collaboration on interdisciplinary research projects can generate a better solution to the problem under study. Universities are exemplars for other institutions and society. Successful sustainable initiatives on campuses can be extended in making the neighbouring communities sustainable. Colleges have revised their curriculum by including research projects that involve the community (Ollervides and Farrell, 2007; Gerritsen *et al.*, 2006; Thomas and Depasquale, 2016). Hansen and Lehmann (2006) found universities that encouraged and promoted joint research, linking students with industry and community proved to be successful in terms of mutual benefit. Despite the importance of sustainability in academic research and its contribution to solving societal problems, the multi-interpretability of sustainability concept continues to pose challenges for research on sustainability (Huge *et al.*, 2016). Beringer and Adomßent (2008) pointed out that HEIs shrug from research when they become the *locus* and subject of study. Ávila *et al.* (2017) argued that despite HEIs being the creators of knowledge, there is little impetus given to innovation, which the authors pointed, is because of low commitment from the administration, lack of technology and limited resources of funds for research.

2.1.6 Sustainability leadership. According to Hargreaves (2007), sustainable leadership underlies the following principles; sustainable leadership promotes and nourishes lifelong learning, lasts and spreads beyond individuals, shares decision-making and believes in social justice. Sustainable leaders do not deplete resources and believe in honouring and revising past organizational memories to create a better future. A sustainable leader is the one who regardless of the leadership position seeks for a sustainable change in the organization (Ferdig, 2007). Cherkowski (2010) reiterated that HEIs should create conditions and opportunities for leadership capacity building among faculty members. Complex and intertwined challenges of sustainability faced by higher education call for an integrative and transformative leadership that practices participatory decision-making and engages the university community and stakeholders in making the university sustainable (Mader *et al.*, 2013). Burns *et al.* (2015) affirmed twenty-first century problems need a new understanding of sustainable leadership that is collaborative, reflective and rooted in the living process paradigm. Scott *et al.* (2012) recommended involving senior leaders in organizational activities and decision-making, selecting leaders with the right capabilities, regular reviewing of the processes, procedures and policies of the leaders and identifying productive approaches to sustainable leadership as some of the critical areas that organizations should work on.

2.1.7 Media. According to Agenda 21, higher education is responsible for promoting sustainability education, training and public awareness. Media can serve as a powerful platform in reaching out to the masses in a fast and effective manner. Media can help create awareness regarding sustainable and unsustainable actions and promote pro-social and environmental conscience. Bodt (2007) reported the efficiency of mass media in reaching out

to the public and in promoting sustainable practices. Media can be used to invite public opinion on unsustainable issues and ways of solving them. A study by [Carpenter et al. \(2016\)](#) revealed that social media is used by educators to spread information, but its usage for other purposes is limited. To capitalize on the reach of social media, institutions should invest in training educators regarding the effective ways of using social media. Social media can support teaching-learning, delivering sustainable education, collaborating with others across campus and geographical boundaries and sharing sustainable campus practices with students and society via social media sites ([Hamid et al., 2017](#)).

2.1.8 Institutional commitment. The proclamation of 2005–2014 as the DESD saw an upsurge in the number of professional associations, charters and declarations signed by HEIs committing to education for sustainability ([Sherman, 2008](#)). [Clugston and Calder \(1999\)](#) stressed that commitment towards sustainability should be reflected in the institute's daily operations, such as teaching and learning, research and community engagement activities. It should be visible in the institute's vision and mission statements. [Velazquez et al. \(2006\)](#) designed a sustainable university model that can be used by universities and colleges planning to incorporate sustainability in their activities. Signing declarations does not guarantee sustainability until it is internalized and institutionalized in all the educational activities of higher education ([Lozano, 2006](#)). [Lozano et al. \(2015\)](#) reiterated that institutional framework, educating the educators and sustainable campus activities should become an integral part of the institution's focus.

2.1.9 Accreditation/sustainability audit. Accreditation and sustainability audits can facilitate sustainability integration by HEIs. Accreditation is the process of recognizing an institution, college or university for being qualified of performing a particular activity. [Cooper et al. \(2014\)](#) observed that accreditation agencies can help organizations adopt sustainability when it is reciprocated with the willingness to change by the people in the organization. [Murray and Murray \(2007\)](#) reiterated that accrediting bodies can influence professional colleges to embrace sustainability. A study by [Nicholls et al. \(2013\)](#) found that business schools accredited by the Association to Advance Collegiate Schools of Business (AACSB) have incorporated ethics, corporate social responsibility and sustainability in their curriculum. A sustainability audit takes into consideration the policies, procedures and practices of an institute and compares it with the best practice. It helps establish standards that the institutions should focus on and track and measure whether these standards are met by the institution. Today a number of assessment tools are available to assess sustainability in higher education. According to [Shriberg \(2002\)](#), the cross-institutional assessment helps identify and benchmark the best practices, communicate goals and provide direction to the institutions in becoming sustainable campuses. In addition, the author reviewed 11 cross-institutional assessment tools and found cross-functional efforts, decreasing consumption of energy, water and other resources as critical parameters in achieving sustainability in higher education. A study by [Saadatian et al. \(2011\)](#) found the sustainability tracking and assessment rating system (STARS) and the campus sustainability assessment framework as the most comprehensive assessment tools. [Sayed and Asmuss \(2013\)](#) also found STARS as the best tool for assessing sustainability in all areas of campus life. STAR was the only tool that covered all five aspects of sustainability in universities, including management, academia, environment, engagement and innovation ([Alghamdi et al., 2017](#)). [Lambrechts and Ceulemans \(2013\)](#) argued that although the All India Survey on Higher Education (AISHE) 1.2 is one of the reliable tools for assessing sustainability in higher education, it is not comprehensive as it lacks parameters to assess research, campus operations and outreach programmes. According to [Berzosa et al. \(2017\)](#), AISHE is an effective tool for assessing a programme, faculty or a department, but is not suitable for overall assessment.

Bullock and Wilder (2016) suggested the global reporting initiative as a comprehensive and valid tool for assessing sustainability in higher education. Yarime and Tanaka (2012) recommended multiple approaches to assessing sustainability as most of the tools lack comprehensiveness. Fischer *et al.* (2015) mentioned the use of environmental management systems that involve external auditing and certification mechanism. Gómez Vecchio (2012) proposed the adaptive model of sustainability assessment that enables local calibration without diluting the relative importance of key sustainability issues. It can recalibrate the existing tools and can serve as a model for institutions that have had problems adapting tools that fit their campus specifications. Caeiro *et al.* (2020) discussed the need to define the objectives of assessment tools and to develop tools that integrate the external impact of HEIs on sustainability, participatory process and non-traditional aspects of sustainability. HEIs carry the responsibility to communicate their actions towards sustainable development to the stakeholders in the form of a sustainability report (Ceulemans *et al.*, 2015). Sassen and Azizi (2018) assessed sustainability reports of US universities and found it to be still in the nascent stages, with the focus of these reports primarily restricted to the environmental dimension. A systematic literature review revealed a fragmented approach to sustainability reporting in higher education (Ceulemans *et al.*, 2015).

2.1.10 Government policies. Policies with a clear focus on knowledge, skills, perspectives and values, funding support and partnership with academia can foster embodying and embedding sustainable development in higher education. Chhokar (2010) discussed the initiatives taken by the Indian Government in incorporating education for sustainable development (ESD) by making environmental studies a compulsory subject at the undergraduate level and encouraging sustainability initiatives focusing on rural development. Government regulations and stringent enforcement of these regulations can foster the infusion of sustainability-focused policy and practices in HEIs (Velazquez *et al.*, 2005). Nomura and Abe (2010) revealed how the government policies, programmes and funding of research projects in sustainability have proved to be the driving force in triggering ESD initiatives in Japan. The Taiwan Sustainable Campus Programme, a government initiative, funds projects that aim to facilitate sustainable campus practices, develop sustainability integrated curriculum and teaching material, trains educators in incorporating sustainability in teaching-learning and networking with the local community in activities related to sustainability (Su and Chang, 2010). China's agenda 21 focuses on educating and capacity building of educators and administrators (Niu *et al.*, 2010). The Australian Government has a number of policies that are aimed at transforming the higher education sector towards sustainability (Ferreira and Tilbury, 2012). Fiselier *et al.* (2018) explained the role of the quality assurance agency and the higher education academy, an initiative by the UK Government in guiding HEIs in integrating sustainability and also in organizing meetings, workshops and conferences, bringing stakeholders together for communicating and initiating new projects.

2.2 Literature review on total interpretive structural modelling approach

Several methods have been used by researchers to identify and understand the relationship between the enablers. One of the widely used and effective methods for assessing the interrelationship among different elements and the degree of association between them is interpretive structural modelling (ISM) methodology, proposed by Warfield (1974). The method is used to crystallize ill-structured mental models into well-articulated hierarchically structured conceptual models (Sushil, 2018a, 2018b; Harikannan *et al.*, 2020). The method helps in identifying the enablers and in establishing a contextual relationship between the enablers but fails to explain the causality of the relationship. This shortcoming of the ISM

method is overcome by the TISM (Sushil, 2012). TISM is an extension of the ISM method. It not only helps in establishing a contextual relationship between the enablers but it also captures the expert opinion on the logical interpretation behind the relationship and depicts it in the digraph. TISM also considers the significant transitive links that are removed in the ISM method, resulting in a better structural model. The method is argued to offer a better solution when compared to the structural equation modelling (SEM) technique as it can establish causality with limited data and can be applied to both cross-sectional and longitudinal data (Luo *et al.*, 2018). TISM answers the key research questions – what, how and why of theory building (Sushil, 2012). The multiple advantages of the TISM method have made this alternative research method widely accepted by the researchers in conceptualization and theory building across varied domains (Dubey *et al.*, 2015; Dubey *et al.*, 2017; Menon and Suresh, 2019). A number of researchers have used the TISM method to identify and understand the interrelationship between the enablers. The method was predominantly applied in the manufacturing sector (Sindhvani and Malhotra, 2017), but finally, the method has found grounding in supply chain management (Sandeepa and Chand, 2018), health care (Vaishnavi *et al.*, 2019a), ICT (Prasad *et al.*, 2018) and other fields.

TISM has been used extensively in the education sector. Menon and Suresh (2020b) used the TISM method to analyse the factors that can promote workforce agility in engineering educational institutions in India. The study identified 16 enablers of which management support, rewards and recognition and employee empowerment were found to be crucial in facilitating workforce agility. A similar study to understand the factors influencing organizational agility in higher education was carried out using the TISM method. The study identified leadership, human resource strategies and organizational structure as crucial enablers that can facilitate higher education agility (Menon and Suresh, 2020c). A study by Kashiramka *et al.* (2019) on factors critical for the success of technical educational institutions revealed autonomy and accountability and availability of sustainable funds are the crucial drivers. Sehgal and Nasim (2018) identified technical knowledge, management skills and communication skills as strong employability predictors for graduates seeking employment in the information technology sector. Yeravdekar and Behl (2017) assessed the enablers that can play a crucial role in the branding of private and public management institutions in India. In another study on management institutions, Mahajan *et al.* (2016) used TISM method to analyse factors challenging management education in India. Prasad and Suri (2011) applied TISM to analyse the continuity and change forces driving private higher education. Thus, based on the literature, TISM was found to be an appropriate method for this study.

3. Research methodology

An extensive literature review helped in identifying the enablers that can facilitate sustainability in higher education. After the enablers were identified, five academic experts with more than 15 years of teaching experience were contacted for any revision to the list of enablers. Expert opinion helped in finalizing the list and in developing the model's feasibility and face validity. A total of 40 faculty members were approached, including the five experts contacted initially. Each respondent was interviewed for 40–60 min. A closed-ended questionnaire using a five-point Likert scale was used for collecting data. The study used the TISM approach for analysing the enablers for promoting sustainability in higher education. The steps involved in TISM approach are explained below:

- *Identify and define the enablers*: TISM model begins with identifying and defining the key enablers. An extensive literature review and expert opinion helped in identifying 10 enablers that can facilitate sustainability in higher education. The identified enablers are listed in [Table 1](#).
- *Establish a contextual relationship between enablers*: Pairwise comparison between the factors helped in establishing and comprehending the relationship between the factors. The relationship between the factors is established on the basis of intent, priority, dependence and enhancement of the attributes (e.g. P influences/enhances/changes/alters Q). If enabler P influences Q, then the strength of the relationship is defined as either very strong, strong, moderate, light or no influence. In generating the initial reachability matrix, only those factors that have a very strong or strong influence on one another are considered. The respective cells corresponding to these factors are labelled 1, while all other cells are labelled 0. The initial reachability matrix for sustainability in higher education is presented in [Table 2](#).
- *Develop final reachability matrix and check for transitivity*: The entries in the initial reachability matrix that are denoted by 0 become eligible for the transitivity check. If transitivity is present, the respective cells will be represented by 1* or 1** and if transitivity is not established the cell will be denoted by 0 ([Patil and Suresh, 2019](#); [Vaishnavi et al., 2019b](#); [Vaishnavi and Suresh, 2020](#); [Suresh and Arun Ram Nathan, 2020](#)).

Illustration: First level transitivity is presented by 1*: if $P = Q$; $Q = R$; then $P = R$

Second level transitivity is presented by 1**: if $P = Q$; $Q = R$; $R = S$ then $P = S$

The final reachability matrix for SHE is presented in [Table 3](#).

- *Partition of the final reachability matrix*: Partitioning of the reachability matrix is mainly based on the three sets, i.e. reachability set, antecedent set and intersection set. The row elements of final reachability matrix are presented in the reachability set and column elements are presented in the antecedent set. The common elements in these two sets are presented in the intersection set. In iteration one, all the factors that have the same elements in the reachability set and the intersection set are assigned the first level and are not considered for the next level. The process goes on until all the factors are assigned to different levels. These matrices are presented in [Appendix \(Table A1–A5\)](#).
- *Use the direct and transitive links to obtain the interaction matrix*: The interaction matrix is designed using the direct links and the significant transitive links. All the cells with such links are coded with the binary value 1 and all the other cells that have no connections are coded with the binary value 0. This is presented in [Table 4](#).
- *Develop a digraph*: The final step is creating a digraph, an acronym for a directed graph. The factors and the interrelationship between the factors are presented in a graphical form using nodes and links. Nodes represent the elements and the link between the factors is represented by unidirectional or bidirectional arrows. At this stage, only the significant transitive links are retained. The factors at the top of the model are called the first level factors. Subsequently, coming down the digraph the numeric value of each level increases in the ascending order, one at a time. The TISM digraph model is presented in [Figure 1](#) and the interpretive–interaction matrix, which offers qualitative notes on the direct as well as the transitive links, is shown in [Table 5](#).

Sl. no.	Enabler	References
1	Sustainability incorporated curriculum	Ceulemans and De Prins (2010); Dagiliūtė and Liobikienė (2015); Roure <i>et al.</i> (2018), Chhokar (2010); Maruna <i>et al.</i> (2018), Sidiropoulos (2018); Savelyeva and McKenna (2011), Ferrer-Balas <i>et al.</i> (2010); Stubbs (2013), Aktas (2015); Shrivastava (2010), Sipos <i>et al.</i> (2008); Balsiger <i>et al.</i> (2017), Piasentin and Roberts (2018); Lozano (2010), Fisher and McAdams (2015); Juárez-Nájera <i>et al.</i> (2006); Lidgren <i>et al.</i> (2006), Kurland <i>et al.</i> (2010); Widener <i>et al.</i> (2016), Wu and Shen (2016); Elkington (1997), Rampasso <i>et al.</i> (2018); Remington-Doucette <i>et al.</i> (2013), Mochizuki and Fadeeva (2010); Rusinko (2010), Pandey and Vedak (2009); Barrett <i>et al.</i> (2017), Perera and Hewege (2016); Wyness and Sterling (2015)
2	Pedagogy	Viegas (2016), Fisher and McAdams (2015); Shephard (2008), Sterling and Thomas (2006); Scott (2015), Meyer (2016); Barth <i>et al.</i> (2007), Kerekes and Wetzker (2013); Seatter and Ceulemans (2017), Bhatia and Bhatia (2008); Sterling and Thomas (2006), Montiel <i>et al.</i> (2018); Steiner and Laws (2006), Posch and Steiner (2006); Steiner and Laws (2006), Stauffacher <i>et al.</i> (2006); Fini <i>et al.</i> (2018), Restrepo <i>et al.</i> (2017); Leal Filho <i>et al.</i> (2016), Wiek <i>et al.</i> (2014); Hallinger and Lu (2011), MacVaugh and Norton (2012); Kolb and Kolb (2005), Savage <i>et al.</i> (2015); Rodriguez-Andara <i>et al.</i> (2018), Rios <i>et al.</i> (2018); Albareda-Tiana <i>et al.</i> (2018), Wang <i>et al.</i> (2018); Martin (2015), Kim <i>et al.</i> (2018); Cusick (2009), Wall <i>et al.</i> (2017); Gordon and Thomas (2018), Shephard (2008); Chandler and Teckchandani (2015), Lozano <i>et al.</i> (2017); Ivanaj <i>et al.</i> (2014), Keles (2015); Österlind (2018); Clemens and Hamakawa (2010), Diamond and Irwin (2013); Jääskelä <i>et al.</i> (2017); Altomonte <i>et al.</i> (2016), Roth <i>et al.</i> (2015); Mercer <i>et al.</i> (2017), Vemury <i>et al.</i> (2018); Christie <i>et al.</i> (2013), Tarrant and Thiele (2016)
3	Educators' skills and competencies	Falkenberg and Babiuk (2014), Stir (2006); Biasutti <i>et al.</i> (2018), Bertschy <i>et al.</i> (2013); Nikolic <i>et al.</i> (2017), Lozano-Garcia <i>et al.</i> (2008); Natkin and Kolbe (2016), Sibbel (2009); Pedersen (2017), Fien <i>et al.</i> (2000); Jaspas (2008)
4	Collaborations/partnerships	Lozano-Ros (2003), Bhatia and Bhatia (2008); Moore (2005), Holmberg and Larsson (2017); Vardhan and Shahul Hameed (2012), Caniglia <i>et al.</i> (2017); Dlouha <i>et al.</i> (2018), Gadsby and Bullivant (2011); Yarime <i>et al.</i> (2012)
5	Research and innovation	Wickson <i>et al.</i> (2006), Waas <i>et al.</i> (2010); Bossio <i>et al.</i> (2014), Tormey <i>et al.</i> (2008); Wooltorton <i>et al.</i> (2015), Aktas (2015); Ollervides and Farrell (2007), Gerritsen <i>et al.</i> (2006); Thomas and Depasquale (2016), Hansen and Lehmann (2006); Hüge <i>et al.</i> (2016); Beringer and Adomßent (2008); Ávila <i>et al.</i> (2017)
6	Sustainability leadership	Hargreaves (2007), Ferdig (2007); Cherkowski (2010), Mader <i>et al.</i> (2013); Burns <i>et al.</i> (2015), Scott <i>et al.</i> (2012)
7	Media	Bodt (2007), Carpenter <i>et al.</i> (2016); Hamid <i>et al.</i> (2017)
8	Institutional commitment	Sherman (2008), Clugston and Calder (1999); Velazquez <i>et al.</i> (2006), Lozano (2006); Lozano <i>et al.</i> (2015)
9	Accreditation/sustainability audit	Cooper <i>et al.</i> (2014), Murray and Murray (2007); Nicholls <i>et al.</i> (2013), Shriberg (2002); Saadatian <i>et al.</i> (2011), Sayed and Asmuss (2013); Alghamdi <i>et al.</i> (2017), Lambrechts and Ceulemans (2013); Berzosa <i>et al.</i> (2017), Bullock and Wilder (2016); Yarime and Tanaka (2012), Fischer <i>et al.</i> (2015); Gómez Vecchio (2012); Caeiro <i>et al.</i> (2020), Ceulemans <i>et al.</i> (2015); Sassen and Azizi (2018)
10	Government policies	Chhokar (2010), Velazquez <i>et al.</i> (2005); Nomura and Abe (2010), Su and Chang (2010); Niu <i>et al.</i> (2010), Ferreira and Tilbury (2012); Fiselier <i>et al.</i> (2018)

Table 1.
Identified enablers
for ESD in higher
education

4. Results

4.1 Interpretation of total interpretive structural modelling digraph

This study aims at identifying the factors that can facilitate sustainability in higher education. A total of 10 factors were identified and were assigned hierarchical levels depending on the interaction matrix given in Table 4. The factors are then arranged in a

hierarchical structure in the TISM digraph model, as presented in Figure 1. The factors placed at the top of the TISM model are assigned the first level. These factors influence each other and have no impact on other factors, whereas all the other factors below the first level can impact factors at their level and the ones above but can get impacted only by the factors that are below them in the structure (Jain *et al.*, 2018). The arrows in the digraph relate to the direction of the relationship between the factors and the dotted lines represent the significant transitive links. The interpretation of the relationship between the factors is discussed below.

Table 2.
Initial reachability
matrix for
sustainability in
higher education

Enablers	E1	E2	E3	E4	E5	E6	E7	E8	E9	E10
E1	1	1	1	1	1	0	0	0	0	0
E2	1	1	1	1	1	0	0	0	0	0
E3	1	1	1	1	1	0	0	0	0	0
E4	1	1	1	1	1	0	0	0	0	0
E5	1	1	1	1	1	0	0	0	0	0
E6	1	1	1	1	1	1	0	1	0	0
E7	0	0	0	0	1	1	1	0	1	0
E8	1	1	1	1	1	1	0	1	0	0
E9	1	0	0	0	0	0	0	1	1	0
E10	1	0	1	0	0	0	1	1	1	1

Table 3.
Final reachability
matrix for
sustainability in
higher education

Enablers	E1	E2	E3	E4	E5	E6	E7	E8	E9	E10
E1	1	1	1	1	1	0	0	0	0	0
E2	1	1	1	1	1	0	0	0	0	0
E3	1	1	1	1	1	0	0	0	0	0
E4	1	1	1	1	1	0	0	0	0	0
E5	1	1	1	1	1	0	0	0	0	0
E6	1	1	1	1	1	1	0	1	0	0
E7	1*	1*	1*	1*	1	1	1	1*	1	0
E8	1	1	1	1	1	1	0	1	0	0
E9	1	1*	1*	1*	1*	1*	0	1	1	0
E10	1	1*	1	1*	1*	1*	1	1	1	1

Table 4.
Interaction matrix

Enablers	E1	E2	E3	E4	E5	E6	E7	E8	E9	E10
Sustainability incorporated curriculum (E1)	1	1	1	1	1	0	0	0	0	0
Pedagogy (E2)	1	1	1	1	1	0	0	0	0	0
Educators' skills and competencies (E3)	1	1	1	1	1	0	0	0	0	0
Collaborations/partnerships (E4)	1	1	1	1	1	0	0	0	0	0
Research and innovation (E5)	1	1	1	1	1	0	0	0	0	0
Sustainability leadership (E6)	1	1	1	1	1	1	0	1	0	0
Media (E7)	0	0	0	0	1	1	1	0	1	0
Institutional commitment (E8)	1	1	1	1	1	1	0	1	0	0
Accreditation/sustainability audit (E9)	1	0	0	0	0	0	0	1	1	0
Government policies (E10)	1	0	1	0	0	0	1	1	1	1

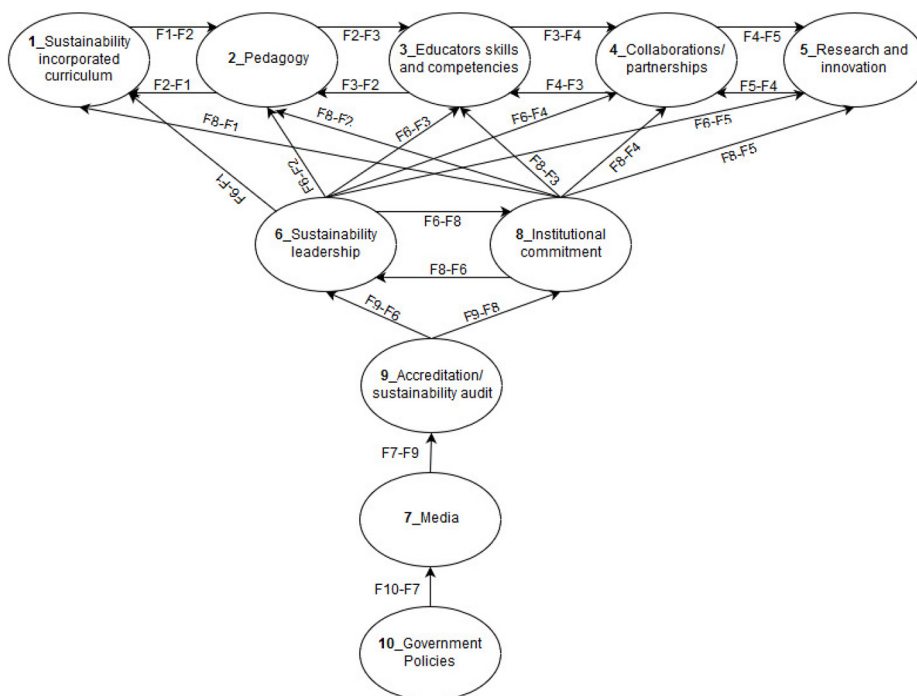


Figure 1.
TISM model for
sustainability in HEIs

4.1.1 Level V: government policies (Enabler 10). Government policy occupies this level and this makes the enabler most influential. According to the interaction matrix, government policies influence sustainability curriculum, educators' skills, media, institutional commitment and accreditation/sustainability audit. Policies, rules and guidelines by the Department of Higher Education and universities can help HEIs in incorporating sustainability in their educational activities. Capacity building initiatives by the government can help educate and train faculty members, empowering them to teach sustainability (Niu *et al.*, 2010). Higher education regulatory bodies and accreditation agencies regularly publish the new/revised rules and regulations on their websites. These changes are adopted by the institutions, fostering the integration of sustainability principles into the system.

4.1.2 Level IV: media (Enabler 7). The next enabler that has a high driving force is media. Media can influence research, sustainability leadership and accreditation/sustainability audits.

Media in general and social media, in particular, serves as a platform for new information and knowledge. Social media sites such as LinkedIn, Research Gate, Twitter and Facebook provide a platform for students, research scholars, educators and industry experts to collaborate and get ideas on currently relevant research topics. This can help the researchers get some quality inputs, engage in meaningful discussions, exchange knowledge and information across disciplines, sectors and geographical boundaries. Social media is also used by researchers to keep them updated on the upcoming workshops, conferences and related events. Social media is a popular platform among faculty members to network with the peer. Cross-pollination of ideas, information, recent developments and sustainable practices adopted by institutions can trigger institutions to take up leadership roles in

Table 5.
Interpretive-
interaction matrix

	E1	E2	E3	E4	E5	E6	E7	E8	E9	E10
E1		Sustainability incorporated curriculum influences pedagogy	Sustainability incorporated curriculum influences educators' skills and competencies	Sustainability incorporated curriculum influences collaborations	Sustainability incorporated curriculum influences research and innovation					
E2	Pedagogy influences sustainability incorporated curriculum		Pedagogy influences educators' skills and competencies	Pedagogy influences collaborations/ partnerships	Pedagogy influences research and innovation					
E3	Educators skills and competencies influence sustainability incorporated curriculum	Educators skills and competencies influence pedagogy		Educators skills and competencies influence collaborations/ partnerships	Educators skills and competencies influence research and innovation					
E4	Collaborations/ partnerships influence sustainability incorporated curriculum	Collaborations/ partnerships influence pedagogy	Collaborations/ partnerships networks influence educators skills and competencies		Collaborations/ partnerships influence research and innovation					
E5	Research and innovation influences sustainability incorporated curriculum	Research and innovation influences pedagogy	Research and innovation influences educators skills and knowledge	Research and innovation influences collaborations/ partnerships						

(continued)

	E1	E2	E3	E4	E5	E6	E7	E8	E9	E10
E6	Sustainability leadership influences sustainability incorporated curriculum	Sustainability leadership influences pedagogy	Sustainability leadership influences educators' skills and competencies	Sustainability leadership influences collaborations/ partnerships	Sustainability leadership influences research and innovation			Sustainability leadership influences institutional commitment		
E7					Media influences research and innovation	Media influences sustainability leadership			Media influences accreditation/ sustainability audit	
E8	Institutional commitment influences sustainability incorporated curriculum	Institutional commitment influences pedagogy	Institutional commitment influences educators' skills and competencies	Institutional commitment influences collaborations/ partnerships	Institutional commitment influences research and innovation	Institutional commitment influences sustainability leadership				
E9	Accreditation/ sustainability audit Influences sustainability incorporated curriculum							Accreditation/ sustainability audit influences institutional commitment		
E10	Government policies influence sustainability incorporated curriculum		Government policies influence educators skills and competencies				Government policies influence media	Government policies influence institutional commitment	Government policies influence accreditation/ sustainability audit	

Table 5.

facilitating ESD. Media is a conduit between the policymakers and society. Media acts as a mouthpiece of the common man and the issues faced by them. Media can influence government bodies and agencies regulating the HEIs to make reforms to the existing system and policies to promote ESD among students and institutions.

4.1.3 Level III: accreditation/sustainability audit (Enabler 9). The enabler at this level is an accreditation/sustainability audit. This enabler influences sustainability curriculum and institutional commitment. Accreditation bodies can influence and trigger the incorporation of sustainability in the curriculum. Management institutions accredited by AACSB have incorporated courses on ethics, corporate social responsibility and sustainability into their curriculum (Nicholls *et al.*, 2013). The use of comprehensive sustainability assessment tools on a regular basis can help institutions keep track of how far they have been successful in implementing sustainability in their practices and processes.

4.1.4 Level II: sustainability leadership and institutional commitment (Enablers 6 and 8). Sustainability leadership influences sustainability curriculum, pedagogy, educators' skills, collaborations, research and institutional commitment.

Sustainability advocates/leaders are ones who believe in sustainability and are willing to embrace the change as well as help others in doing so. Such leaders, by providing skill development programmes, opportunities for personal and professional development, can enhance educators' skills in designing and developing a curriculum that incorporates principles of sustainability and align it with the right and appropriate pedagogical tools. Faculty learning communities and the community of practice can help enhance collegiality and collaboration between the faculty members and provide a platform for them to discuss and reflect on new teaching practices and other related matters (Nikolic *et al.*, 2017). By incorporating novel appraisal and reward systems, leaders can motivate faculty members to engage in sustainability-related research. Institutional policies, procedures and practices that integrate principles and values of sustainability show the inclination of the management leaders towards sustainability.

The other enabler at this level is institutional commitment. Institutional commitment influences sustainability curriculum, pedagogy, educators' skills, collaborations, research and innovation and sustainability leadership.

HEIs committed to sustainability integrate values, activities and principles of sustainability into all aspects of education and learning. Adopting sustainability begins by incorporating sustainability into academic content. The new curriculum brings in new and innovative teaching techniques that take students closer to reality, enhancing their ability to understand the complex issues of sustainability. Faculty members' beliefs about teaching and learning are socially constructed from their local workplaces, department, university and the research community that they are a part of. An institution that harbours an organizational culture that envisions a sustainable future will be effective in transforming the beliefs of the stakeholders, especially the faculty members, by providing them with opportunities to enhance their knowledge and abilities. Institutions can encourage research in sustainability by acknowledging and rewarding such projects and by funding sustainability research. HEIs can also provide for leadership development opportunities that enable and empower the leaders to identify successful solutions to the problems faced by them in implementing changes as part of the transformation of college or university to a sustainable university or college.

4.1.5 Level I: sustainability incorporated curriculum, pedagogy, educators' skills and competencies, collaborations and research (Enablers 1, 2, 3, 4 and 5). This level consists of five variables. The enablers at this level impact each other and get impacted by other enablers in the system. Sustainability incorporated curriculum (E1) influences pedagogy, educators'

skills, collaborations and research. The integration of sustainability principles and values into the curriculum calls for the adoption of a renewed pedagogical approach. A change in the curriculum and new teaching practices require educators to acquire new skills and competencies. Interdisciplinary courses/programmes influence collaboration between students and faculty members of different disciplines (Widener *et al.*, 2016). Sustainability-focused programmes encourage research and innovation as students work in collaboration with the community in solving sustainability issues faced by them (Wickson *et al.*, 2006).

Pedagogy (E2) impacts the sustainability curriculum, educators' skills, collaborations and research. ESD should empower the students to make informed decisions that are environmentally, socially and economically viable. Teaching a value-laden subject like sustainability requires the faculty members to practice and believe in sustainability. An empowered teacher can be instrumental in bringing about a change in the values and belief system of students by incorporating sustainability in the academic content and aligning it with innovative teaching practices. Understanding the sustainability concept and different pedagogical approaches can influence faculty members to develop the better sustainability-focused syllabus. A teacher needs to be well-versed with the new teaching methods and this may require upskilling educators' skills. Faculty learning communities and development programmes can help faculty members discuss and reflect on new teaching practices. Learning new teaching approaches can influence research on innovative ways of teaching sustainability.

Educators' skills and competencies (E3) influence sustainability curriculum, pedagogy, collaborations and research. Educators' when equipped with the knowledge and skills required to develop a syllabus on sustainability, can influence curriculum and teaching practices. Educators involved in research of new instructional ways of teaching sustainability can design and develop new pedagogical tools. Educators who are comfortable with the concept of sustainability will seek collaboration with other educators and researchers to keep themselves updated on the latest in education for sustainability. To find solutions to the complex issues faced by society, it is important that educators across boundaries collaborate, deliberate and research to come out with practical solutions to such complex issues.

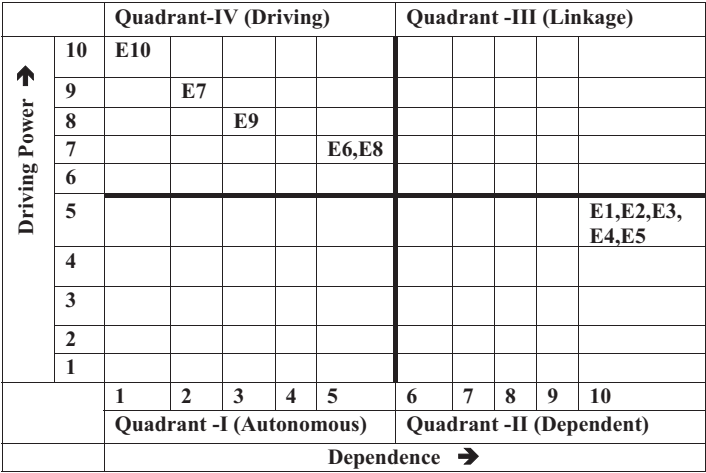
Collaborations (E4) influence sustainability curriculum, pedagogy, educators' skills and research. Collaborating with a peer group, researchers and experts can better the understanding of educators regarding sustainability concept and this would help them design and develop academic content on sustainability and incorporate innovative teaching practices. Collaborations can necessitate mutual learning, openness, understanding and respect for others and help educators improve their social competency. Collaborations can influence educators, researchers and institutions to come together and search for practical solutions to real-world sustainability issues.

Research and innovation (E5) influence sustainability curriculum, pedagogy, educators' skills and collaborations. Action research cycles can help in curriculum development. Research can lead to the development of new instructional methods. Sustainability issues are not localized and to find solutions to the problems, research studies need to collaborate across boundaries and disciplines. Interdisciplinary/trans-disciplinary research can enhance the knowledge, skills and ability of the educators.

4.2 Cross-impact multiplication matrix analysis

MICMAC analysis helps in analysing and categorizing enablers (Lakshmi Priyadarsini and Suresh, 2020). Variables are categorized into four quadrants – driving, dependent, linkage and autonomous, depending on their driving and dependence power (Suresh *et al.*, 2019a; Suresh *et al.*, 2019b) (Figure 2).

Figure 2.
Diagram of MICMAC
analyses



4.2.1 Interpretation of cross-impact multiplication matrix analysis. The first quadrant is the group of *autonomous enablers* that have weak driving power and weak dependence power. In this diagram, there are no autonomous enablers.

The second quadrant is the group of *dependent enablers* that have a weak driving power but a strong dependence power. The dependent enablers in this study are sustainability incorporated curriculum (E1), pedagogy (E2), educators' skills and competencies (E3), collaborations/partnerships (E4) and research and innovation (E5). These enablers are dependent on enablers with the high driving force for activation.

The third quadrant consists of *linkage enablers* that have a high driving and dependence power. There are no linkage enablers in this study.

The fourth quadrant consists of a group of *driving enablers* that have a strong driving power but a weak dependence power. In this study, the driving enablers are government policies (E10), media (E7), accreditation/sustainability audit (E9), sustainability leadership (E6) and institutional commitment (E8). The driving enablers are the key enablers.

The MICMAC ranking of identifying enablers for sustainability in higher education is mentioned in Table 6.

Table 6.
MICMAC analysis
rank for ESD

Enablers	Driving power	Dependence	Driving power/Dependence	MICMAC rank
E1	5	10	0.500	5
E2	5	10	0.500	5
E3	5	10	0.500	5
E4	5	10	0.500	5
E5	5	10	0.500	5
E6	7	5	1.400	4
E7	9	2	4.500	2
E8	7	5	1.400	4
E9	8	3	2.667	3
E10	10	1	10.000	1

4.3 Total interpretive structural modelling validation

The links between the enablers were further validated by conducting a pre-phase quantitative analysis. The response of 12 respondents with more than 10 years of experience was selected for validating 42 links, taken from the interaction matrix. As mentioned in the research methodology, the responses were collected using a five-point Likert scale ranging from “no influence” to “very strong influence”.

4.3.1 Research hypotheses.

H₀. There is no significant difference between the observed mean and specified mean in relation to the experts’ opinion.

H_A. There is a positive significant difference between the observed mean and specified mean in relation to the experts’ opinion.

i.e. *H₀*: mean (observed) – mean (specified) = 0

H_A: mean (observed) – mean (specified) > 0

Here, degree of freedom = 12 – 1 = 11

A one-tailed one-sample *t*-test was found appropriate to test the hypotheses (Menon and Suresh, 2020c). The “hypothetical mean value” was set as 2.6. Microsoft Excel was used to run the test. The results of hypothesis testing are shown in Table 7. At a 5% significance level, all those links that had a value of less than 0.025 (significance value for a one-tailed test) were accepted and other links were rejected (Yadav, 2014). Out of 42 links, the majority of the links were accepted, and only 7 links got rejected.

5. Theoretical and managerial implications

5.1 Theoretical implications

According to Whetten (1989), any research should address the fundamental questions: *what, how, why, where* and *when*. This study attempts to provide a theoretical contribution to the existing literature on sustainability integration in higher education by answering the key research questions of theory building *what, how* and *why*. Sushil (2012) posited that the TISM method helps in identifying the enablers that can promote sustainability integration in higher education (what), establish a relationship between the enablers’ (how) and also provides a logic interpretation that explains the causality of the relationship (why). This study contributes to the existing literature on sustainability in higher education by providing a comprehensive list of factors and establishing a multi-directional interplay between the factors.

5.2 Managerial implications

The structured model has incorporated and prioritized all the crucial drivers that would help the top management of colleges and universities in higher education to adopt strategies that can facilitate sustainability. Reformulating strategies, processes and practices by focusing on variables with a strong driving force can promote ESD. Sustainability should become a part of the institute’s culture and be reflected in the institute’s vision, structure, activities and daily operations. Institutions should enable and empower sustainability leaders and steering committee members to explore, develop and implement sustainability initiatives. It is important that HEIs adopt practices such as training and development of faculty members and incentivizing sustainable initiatives and research. Sharing sustainable practices on campus via social media can help other institutions and the extension of these practices can help the neighbouring communities.

Table 7.
Results of
hypotheses testing

Sl. no.	Enablers linked	Mean	SD	t-Value	Sig. (1-tailed)	Accept/reject
E1-E2	Sustainability curriculum influences pedagogy	4.083333	0.514929	9.978892	0.00000037	Accept
E1-E3	Sustainability curriculum influences educators' skills and competencies	3.666667	1.073087	3.443377	0.002745926	Accept
E1-E4	Sustainability curriculum influences collaborations	3.25	1.215431	1.852566	0.045472281	Reject
E1-E5	Sustainability curriculum influences research and innovation	3.416667	0.792961	3.567659	0.002206573	Accept
E2-E1	Pedagogy influences sustainability curriculum	2.666667	1.230915	0.187617	0.427296587	Reject
E2-E3	Pedagogy influences educators' skills and competencies	3.75	0.965307	4.12689	0.000840477	Accept
E2-E4	Pedagogy influences collaborations	2.666667	1.302678	0.177281	0.431254921	Reject
E2-E5	Pedagogy influences research and innovation	3.583333	1.083625	3.143493	0.004675	Accept
E3-E1	Educators' skills and competencies influence sustainability curriculum	3.5	1	3.117691	0.00489511	Accept
E3-E2	Educators' skills and competencies influence pedagogy	4	0.953463	5.086453	0.000175707	Accept
E3-E4	Educators' skills and competencies influence collaborations	3.5	0.797724	3.908233	0.001220872	Accept
E3-E5	Educators' skills and competencies influence research and innovation	3.75	0.753778	5.284998	0.000129146	Accept
E4-E1	Collaborations influence sustainability curriculum	3.333333	1.230915	2.063783	0.031725017	Reject
E4-E2	Collaborations influence pedagogy	3.583333	1.083625	3.143493	0.004675	Accept
E4-E3	Collaborations influence educators' skills and competencies	4.166667	0.57735	9.4	0.00000068	Accept
E4-E5	Collaborations influence research and innovation	3.5	1	3.117691	0.00489511	Accept
E5-E1	Research and innovation influences sustainability curriculum	3.833333	0.834847	5.117574	0.000167367	Accept
E5-E2	Research and innovation influences pedagogy	4.083333	0.792961	6.480034	0.000022	Accept
E5-E3	Research and innovation influences educators' skills and competencies	3.5	0.904534	3.446738	0.00272969	Accept
E5-E4	Research and innovation influences collaborations	3.833333	0.57735	7.4	0.0000067	Accept
E6-E1	Sustainability leadership influences sustainability curriculum	3.5	0.797724	3.908233	0.001220872	Accept
E6-E2	Sustainability leadership influences pedagogy	3.583333	0.900337	3.783437	0.001514488	Accept
E6-E3	Sustainability leadership influences educators' skills and competencies	4.083333	0.514929	9.978892	0.00000037	Accept
E6-E4	Sustainability leadership influences collaborations	3.583333	0.900337	3.783437	0.001514488	Accept
E6-E5	Sustainability leadership influences research and innovation	4	0.852803	5.686827	0.000070	Accept
E6-E8	Sustainability leadership influences institutional commitment	3.166667	1.193416	1.64485	0.064123216	Reject
E7-E5	Media influences research and innovation	3.416667	0.996205	2.839794	0.008043906	Accept
E7-E6	Media influences sustainability leadership	2.75	0.621582	0.835957	0.210483588	Reject
E7-E9	Media influences accreditation/sustainability audit	3.166667	1.029857	1.90608	0.041545299	Reject

(continued)

Table 7.

Sl. no.	Enablers linked	Mean	SD	t-Value	Sig. (1-tailed)	Accept/reject
E8-E1	Institutional commitment influences sustainability curriculum	3.75	0.965307	4.12689	0.000840477	Accept
E8-E2	Institutional commitment influences pedagogy	3.833333	1.029857	4.148528	0.000810242	Accept
E8-E3	Institutional commitment influences educators' skills and competencies	3.583333	0.792961	4.295753	0.000632458	Accept
E8-E4	Institutional commitment influences collaborations	4.083333	0.900337	5.707218	0.000068	Accept
E8-E5	Institutional commitment influences research and innovation	3.333333	0.887625	2.861952	0.007731276	Accept
E8-E6	Institutional commitment influences sustainability leadership	3.166667	0.834847	2.351318	0.01919894	Accept
E9-E1	Accreditation/sustainability audit influences sustainability curriculum	3.25	0.965307	2.33259	0.019843737	Accept
E9-E8	Accreditation/sustainability audit influences institutional commitment	4.083333	0.792961	6.480034	0.000022	Accept
E10-E1	Government policies influence sustainability curriculum	4	0.603023	8.042388	0.0000031	Accept
E10-E3	Government policies influence educators' skills and competencies	3.916667	1.083625	4.209084	0.000731511	Accept
E10-E7	Government policies influence media	3.583333	0.900337	3.783437	0.001514488	Accept
E10-E8	Government policies influence institutional commitment	4.25	0.866025	6.6	0.000019	Accept
E10-E9	Government policies influence accreditation/sustainability audit	3.666667	0.778499	4.746367	0.000301607	Accept

6. Conclusion

According to the MICMAC analysis, government policies are found to have the highest driving power and this makes government policies a strong enabler, which can drive sustainability integration in higher education. Apart from government policies, the other enablers with high driving force are media, accreditation/sustainability audit, institutional commitment and sustainability leadership. These enablers are characterized by a high driving force and low dependence power. This makes the enablers independent and can individually impact other enablers, especially the enablers with high dependence and low driving force. In this study, the sustainability curriculum, pedagogy, educators' skills and research are the dependent enablers. Being low in driving force and high on dependence, these variables are dependent on the driving enablers to activate them as they cannot influence the system on their own. Government policies and regulations with a clear focus on knowledge, skills, perspectives and values that an individual should embody for sustainable development would help universities and colleges align their policies and practices. Rigorous regulations, enforcement and funding by the government can catalyze the adoption of practices that would promote sustainability in higher education. Accreditation agencies and comprehensive sustainability assessment tools can ensure colleges comply with the assessment requirements. Higher education can make effective use of media to disseminate innovative practices and research to the public for promoting sustainable living.

The Government of India has recently initiated a few schemes that would promote sustainability education in India. Scheme for trans-disciplinary research for developing economy, a government initiative aims to promote and support research projects that have social, local, national and international significance (Mathrubhumi, 2020). A framework for eco-friendly and sustainable campus development was started to encourage colleges to incorporate policies and practices that promote environmentally sustainable campus practices and Mulya Pravah are the guidelines to infuse human values and ethics in higher education (Mathrubhumi, 2020). The current National Accreditation and Assessment Council manual emphasize incorporating value-added courses that sensitize students to environmental sustainability issues, gender equity, human values and professional ethics. The manual also emphasizes on student involvement in extension activities so as to facilitate students-community collaboration in solving social issues (NAAC manual, 2020). Recently, the University Grant Commission suggested colleges and universities to include Swachh Bharat Abhiyan activities as an elective course under choice based credit system (Singh, 2018). HEIs have been identified as the key drivers for achieving the SDGs. Adopting policies and practices focusing on sustainable development principles and values in all activities of education and learning, building collaborations with the stakeholders and transparency in governance are some of the issues that the universities need to address to lead the society in achieving SDGs (Kang and Xu, 2018). Initiatives like these by the government, commitment of leaders, comprehensive assessment tools and skilled faculty members can catalyze the process of integration of sustainability in higher education.

This study has certain limitations. The study is restricted to a small sample, which makes generalization difficult. The theoretical framework of the enablers of sustainability in higher education was validated by experts. This prelim qualitative validation can be further empirically tested using statistical techniques such as factor analysis and SEM. The study can be extended by including a large sample size from different geographical regions to get a comprehensive picture of the variables that can promote sustainability in higher education.

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Table A1.
Iteration-1 level
partition matrix

Appendix

Enablers	Reachability set	Antecedent set	Intersection set	Level
1	1, 2,3,4,5	1, 2,3,4,5,6,7,8,9,10	1, 2,3,4,5	I
2	1, 2,3,4,5	1, 2,3,4,5,6,7,8,9,10	1, 2,3,4,5	I
3	1, 2,3,4,5	1, 2,3,4,5,6,7,8,9,10	1, 2,3,4,5	I
4	1, 2,3,4,5	1, 2,3,4,5,6,7,8,9,10	1, 2,3,4,5	I
5	1, 2,3,4,5	1, 2,3,4,5,6,7,8,9,10	1, 2,3,4,5	I
6	1, 2,3,4,5,6,8	6,7,8,9,10	6,8	
7	1, 2,3,4,5,6,7,8,9	7,10	7	
8	1, 2,3,4,5,6,8	6,7,8,9,10	6,8	
9	1, 2,3,4,5,6,8,9	7,9,10	9	
10	1, 2,3,4,5,6,7,8,9,10	10	10	

Table A2.
Iteration-2 level
partition matrix

Enablers	Reachability set	Antecedent set	Intersection set	Level
6	6,8	6,7,8,9,10	6,8	II
7	6,7,8,9	7,10	7	
8	6,8	6,7,8,9,10	6,8	II

Table A3.
Iteration-3 level
partition matrix

Enablers	Reachability set	Antecedent set	Intersection set	Level
7	7,9	7,10	7	
9	9	7,9,10	9	III
10	7,9,10	10	10	

Table A4.
Iteration-4 level
partition matrix

Enablers	Reachability set	Antecedent set	Intersection set	Level
7	7	7,10	7	IV
10	7,10	10	10	

Table A5.
Iteration-5 level
partition matrix

Enabler	Reachability set	Antecedent set	Intersection set	Level
10	10	10	10	V

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